Evidence-Bound Review Under a Fixed Deadline: A Reliability Contract for ICML-Style Review

Abstract

Time-bounded scientific review must combine evidence tracing with reliable batch execution. We implement an anonymous Track 2 review system that keeps platform side effects in a single root coordinator, delegates paper-local drafting to three bounded workers, and validates a canonical review schema before any post attempt. Across three deterministic 10-paper synthetic-mock runs, the runtime completed 30/30 assigned reviews with 100% schema validity and zero duplicate post attempts or idempotency keys; a separate fault run completed 10/10 with schema validity 10/10, no duplicates, one recovery each for malformed JSON output, worker timeout, claim timeout, and post timeout, and one in-process ledger reopen. In an actual two-process dry-run, the first process exited 75 after four mock-verified papers and a fresh process exited 0 at 10/10 with schema validity 10/10 and no duplicates while preserving the four-paper ledger prefix and completed manifest and outbox bytes. On frozen seeded synthetic issues, the deterministic reviewer scored TP 20, FP 0, FN 0, and F1 1.0, compared with TP 10, FP 0, FN 10, and F1 0.6667 for the naive baseline. These measurements are mock-only and do not establish live-platform throughput or review quality on unseen papers.

Serialized platform lane: discover, claim, reconcile, post, verify

↓ frozen manifest and bounded lease

Three review workers: paper-local evidence analysis only

↓ ReviewDraft

Deterministic schema validation and atomic ledger transition

Figure 1. Authority is narrow by construction. Workers cannot produce remote side effects or mutate shared state.

1. Problem and Contract

Track 2 requires an ICML-style review agent and a structured review of a frozen paper. The live operating window is 25 minutes, so review quality and batch reliability are coupled: a well-reasoned draft is not useful if it is lost, duplicated, or posted without confirmation. The system therefore treats evidence identity, review validity, and side-effect state as one end-to-end contract.

The official upstream skill is pinned at commit a9f4f258...153473f. Its Track 2-only path requires no new training, GPU, experiment service, or training metric. It requires frozen paper and evidence hashes, an independently frozen agent definition, a structured result, and explicit reporting when supplied evidence is insufficient. The local wrapper preserves the upstream Contribution assessment while retaining the event fields Significance, Originality, and Comment; it does not convert one field into another.

2. System Design

2.1. Single Owner for Side Effects

A root coordinator is the only component permitted to discover assignments, claim papers, inspect remote status, post reviews, or update the shared ledger. Three persistent review workers receive one frozen paper manifest at a time and return only a structured ReviewDraft. This boundary prevents worker retries from becoming duplicate platform actions.

2.2. Frozen Inputs and Evidence Trace

Before a worker starts, the coordinator records the paper identifier, paper file identity, SHA-256 and size, evidence file identities, hashes and sizes, aggregate input and evidence hashes, agent version, prompt hash,

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and schema hash in a per-paper manifest. The frozen corpus supplies the allowed file paths separately, and Root grants the active lease separately from the manifest. Unsafe paper identifiers are rejected before any artifact or ledger write. A worker must return the frozen identities and its Root-issued lease identity with page, section, table, figure, or saved-result locations for central strengths and weaknesses. An identity mismatch blocks execution and cannot enter the targeted repair path.

2.3. Canonical Review Shape

The draft contains Summary, Strengths, Weaknesses, Questions, Ethics and Limitations, Evidence Trace, and score rationales. It carries Soundness, Presentation, Contribution, Significance, Originality, Overall Recommendation, Confidence, and a constructive Comment as distinct values. A deterministic validator checks required fields, integer ranges, non-empty prose, lease and paper identity, evidence locations, and forbidden filler text. At most one targeted repair may address a schema defect; it may not replace the frozen paper or evidence.

3. Reliability Model

The coordinator uses a bounded queue with a default high-water mark of three, one active lease per paper, append-only ledger writes with atomic snapshot replacement, and idempotency keys for post intents. A timed-out worker may be re-queued once under the bounded retry policy while Root retains the deterministic lease. Malformed output records a validator-reject state and its validation errors before it can enter the single targeted repair path.

Claim and post timeouts are ambiguous side effects. The coordinator therefore queries status before issuing the same action again. A post attempt remains post_unknown until remote state resolves it to verified success or a safe retry. The only live completion condition is posted_verified == assigned_count. Mock completion is reported as mock evidence and never as live platform performance.

4. Evaluation Protocol

Evaluation is split into quality and runtime contracts. Quality cases are seeded with issue type and evidence location before any agent call. A naive single-pass reviewer is frozen as the baseline. Issue matching reports true positives, false positives, and false negatives; location matching follows the predeclared page, section, table, and figure rules. The acceptance threshold is stored with the fixture rather than chosen after observing results.

Table 1. Deterministic synthetic-mock runtime evidence. No production claim or post occurred.

Run	Verified	Schema	Dup.
Normal, three repetitions	30/30	30/30	0
Fault injection	10/10	10/10	0
Two-process resume	10/10	10/10	0

Runtime evaluation uses ten mock assignments and three independent repetitions. It records assigned and verified-complete counts, schema validity, duplicate post count, and latency from raw event records. Separate fault scenarios inject malformed JSON, worker timeout, claim timeout, and post timeout, and reopen the ledger once in-process. Each scenario must show either verified recovery or an explicit unresolved state; silent loss is not a pass.

5. Synthetic Mock Results

Table 1 reports only deterministic synthetic fixtures and the mock adapter. Across three normal runs, all 30 assigned paper instances reached mock posted_verified; all 30 drafts passed the schema, and duplicate post attempts and duplicate idempotency keys were both zero. A separate 10-paper run injected one malformed JSON output, worker timeout, claim timeout, and post timeout, and reopened the ledger once in-process. Each injected condition was recovered once, all ten drafts were valid and mock-verified, and duplicate post attempts remained zero. A no-op rerun after full completion left the ledger and outbox byte-identical.

Actual process recovery was tested separately. One process durably exited with code 75 after four mock-verified papers. A fresh process opened the same output directory and inputs, exited 0 at 10/10 mock verification with schema validity 10/10 and zero duplicate posts or keys, preserved the four-paper ledger prefix, and preserved the completed manifest and outbox bytes.

On the pre-frozen seeded synthetic issue set, the deterministic reviewer found 20 true positives, zero false positives, and zero false negatives, yielding F1 1.0. The naive single-pass baseline found 10 true positives, zero false positives, and 10 false negatives, yielding F1 0.6667. Both had location accuracy 1.0 under the frozen location rule. These results test deterministic fixture recognition, not scientific judgment or generalization.

Table 2. Seeded synthetic-issue quality evaluation.

Reviewer	TP	FP	FN	F1
Deterministic runtime	20	0	0	1.0000
Naive single pass	10	0	10	0.6667

Runtime and quality values come from the frozen evidence/mock-validation-final-20260712T1335KST/ aggregate.json. The actual process result comes from evidence/process-restart-proof/aggregate.json. Values above are direct aggregates or four-decimal rounding of their stored values.

6. Limitations

Passing mock runs cannot establish live platform throughput, submission success, review correctness on unseen research, or agreement with human judgment. The runtime elapsed times reflect in-process synthetic fixtures and are intentionally not used as a live speed claim. Worker-timeout evidence uses a mock adapter TimeoutError and bounded recovery; it does not prove that a running thread was forcibly killed.

Further limitations are structural. Evidence locations can show where a claim was derived but cannot guarantee the claim is correct. Originality assessment is limited to the submitted paper and its supplied references. A deterministic schema validator can reject malformed reviews but cannot certify scientific quality. The live adapter remains unspecified until read-only discovery reveals the actual platform contract; no endpoint or selector is assumed here.

7. Conclusion

The design combines frozen evidence identity, a narrow side-effect owner, bounded paper-local workers, deterministic validation, and status-first reconciliation. Its evaluation contract separates review quality from runtime reliability and separates mock evidence from live outcomes. Deterministic mock runs satisfy the frozen runtime, seeded-fixture, and two-process resume checks, while live platform behavior and unseen-paper review quality remain unmeasured.

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References

Ralphthon ICML organizers. Auto Research Skill, Track 2-only frozen-paper path. Repository commit a9f4f2583648ef4ca54f980f951ae393d153473f, 2026.

Ralphthon ICML organizers. Track 2 rules, 2026.